

Real-Time Phrase-Level Sign Language Recognition Using Bidirectional LSTM Networks over MediaPipe Holistic Keypoint Sequences

Anshul Rehpade, Venkatesh Mudaliar, Rajanee Shenkar

MS in Data Science
Stevens Institute of Technology
Hoboken, NJ

Abstract: Sign language is a primary mode of communication for individuals with speech and hearing impairments, yet most automatic recognition systems classify static hand poses or isolated signs from single video frames and therefore ignore the temporal motion patterns that give continuous sign phrases their meaning. This paper presents a real-time, on-device system for recognizing dynamic, phrase-level sign gestures and converting them into text and synthesized speech. Because existing benchmark datasets such as WLASL do not contain the specific phrase vocabulary required for this task, we construct a custom dataset of 1,500 video clips spanning six classes: *Hello*, *Thank You*, *Yes*, *No*, *Peace*, and a dedicated *None* class that captures everyday non-signing motions. Each video frame is converted into a 258-dimensional pose-and-hand keypoint vector using MediaPipe Holistic, and the resulting temporal sequences are used to train a Bidirectional Long Short-Term Memory (Bi-LSTM) network with TimeDistributed dense layers, dropout regularization, and class weighting. A rolling-window inference pipeline that combines frame smoothing, confidence thresholding, and a label-stability rule enables stable predictions at 20–30 frames per second. The final model achieves **85.57% test accuracy** across the six gesture classes, demonstrating the feasibility of lightweight, privacy-preserving, phrase-level sign language recognition on consumer hardware.

Index Terms: sign language recognition, Bidirectional LSTM, MediaPipe Holistic, gesture recognition, real-time inference, on-device deep learning, accessibility.

1 Introduction

Sign language is the primary means of communication for millions of people with speech and hearing impairments, yet the absence of widespread, automatic translation tools continues to limit interaction between sign language users and the broader hearing community. A real-time system that observes a live camera feed and converts dynamic sign phrases into text or spoken audio could substantially reduce this communication gap, particularly if the system runs entirely on-device, preserving user privacy and avoiding network latency.

Most existing automatic sign language recognition approaches, however, are built around static images or per-frame classification. These approaches treat each video frame independently and therefore cannot capture the motion trajectories that distinguish one sign phrase from another. Many sign phrases share very similar hand shapes when viewed as isolated frames and are only distinguishable through the way the hands, arms, and body move *over time*. A recognition system that ignores this temporal structure is fundamentally mismatched to the task.

A second obstacle is the scarcity of datasets that match the specific phrase-level vocabulary required for a practical assistive system. Large isolated-sign datasets such as WLASL [1] provide a useful benchmark for word-level sign recognition, but their gesture definitions do not align with the continuous, phrase-level gestures targeted in this work, and using them directly leads to high misclassification rates.

To address these challenges, this paper makes the following contributions:

- We construct a **custom phrase-level gesture dataset** of 1,500 video clips across six classes, including a deliberately diverse and “noisy” *None* class that contains everyday

hand movements unrelated to signing.

- We design a **keypoint-based Bi-LSTM architecture** that operates on 258-dimensional MediaPipe Holistic pose-and-hand features extracted from each frame, enabling the model to learn the temporal evolution of a gesture rather than its static appearance.
- We implement a **real-time, on-device inference pipeline** that combines a rolling sequence window, temporal smoothing, confidence thresholding, and a label-stability rule to deliver stable predictions at 20–30 FPS.
- We provide an **empirical evaluation** of the system, including per-class precision/recall/F1, a validation confusion matrix, training/validation learning curves, and a final test accuracy of 85.57% across all six gesture classes.

The remainder of this paper is organized as follows. Section 2 reviews related work on sign language recognition and sequence modeling. Section 3 describes the construction of our custom dataset. Section 4 details the data processing pipeline, baseline models, and the proposed Bi-LSTM architecture. Section 5 describes the real-time inference pipeline. Sections 6 and 7 present the experimental setup and results. Section 8 discusses strengths, limitations, and the role of AI-assisted development, and Section 9 concludes with directions for future work.

2 Related Work

Isolated sign recognition. A large body of work treats sign language recognition as a video classification problem over a fixed vocabulary of isolated signs. The Word-Level American Sign Language (WLASL) dataset [1] is among the most widely used benchmarks of this kind, pairing thousands of short clips with single-word labels and motivating a range

of CNN- and skeleton-based classifiers. While valuable for benchmarking word-level recognition, such datasets are built around a fixed sign vocabulary that does not necessarily reflect the continuous, multi-word phrases required by a conversational assistive system, which motivated us to build a custom dataset for this project.

Continuous and sequence-based recognition. Recognizing continuous sign language [5] requires modeling how hand shape, orientation, and motion evolve across many frames. Recurrent architectures, and in particular Long Short-Term Memory (LSTM) networks [2], were designed to capture exactly this kind of long-range temporal dependency, and bidirectional variants [3] extend this by processing a sequence in both the forward and backward directions, allowing the network to use future as well as past context when interpreting the current frame. These properties make Bi-LSTM networks a natural fit for phrase-level gesture recognition, where the meaning of an early frame in a gesture often depends on how the gesture concludes.

Keypoint and pose-based pipelines. Rather than operating directly on raw pixels, several recent systems first extract structured pose and hand landmarks and then perform sequence classification on the resulting numerical representation. MediaPipe [4] provides an efficient, on-device framework for extracting hand, face, and body-pose landmarks from a live camera feed, which substantially reduces the dimensionality of the recognition problem compared to raw video while preserving the spatial structure of the signer’s body. Our work follows this keypoint-based paradigm, combining MediaPipe Holistic landmark extraction with a Bi-LSTM sequence classifier.

Positioning of this work. Compared to CNN-based frame classifiers and heavier transformer-based sequence models, the approach taken here is intentionally lightweight: a compact Bi-LSTM operating on low-dimensional keypoint sequences, combined with a rule-based stability layer at inference time. This combination is designed specifically for real-time, on-device operation on a small, custom phrase vocabulary, rather than for large-vocabulary offline translation.

3 Dataset Construction

Because no publicly available dataset matched the specific dynamic sign phrases targeted by this project, we built a custom phrase-level gesture dataset from scratch. The dataset consists of **1,500 video clips** spanning six categories: *Hello*, *Thank You*, *Yes*, *No*, *Peace*, and a dedicated *None* class.

Diversity of recordings. Each team member contributed approximately 150–200 recordings, deliberately introducing variation along several axes: camera viewpoint, left- versus right-hand dominance, gesture speed, distance from the camera, and individual performance style. This diversity was intended to help the trained model generalize beyond any single recording setup or signing style, which is critical for a system intended to operate on a wide range of consumer devices and users.

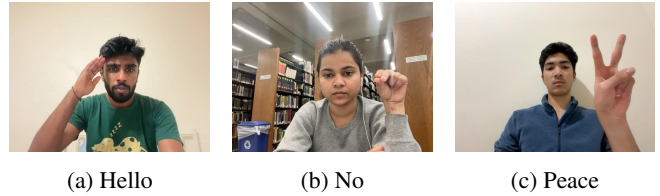


Figure 1: Representative frames from the Hello, No, and Peace gesture classes in the custom dynamic-gesture dataset. The dataset also includes Thank You, Yes, and a dedicated None class that captures everyday non-signing actions rather than idle frames.

The *None* class. A particularly important design decision was to make the *None* class robust rather than trivial. Instead of representing “no gesture” as static, idle frames, this class includes everyday hand movements such as eating, drinking, adjusting clothing, and other ordinary non-signing activities. Including this kind of natural background motion substantially reduced false-positive gesture detections during real-time inference, since the model learns to distinguish deliberate signing motion from incidental hand movement rather than simply from the presence or absence of motion.

Figure 1 shows representative frames from three of the six gesture classes in the custom dataset.

4 Methodology

4.1 Keypoint Extraction with MediaPipe Holistic

Every video clip is processed with the MediaPipe Holistic framework, which extracts hand, body-pose, and facial landmarks from each frame. For the recognition model, each frame is reduced to a fixed-length numerical feature vector consisting of:

- **Pose landmarks:** 33 upper-body landmarks, each with $(x, y, z, \text{visibility})$ coordinates, giving $33 \times 4 = 132$ values.
- **Hand landmarks:** 21 landmarks per hand for both hands, each with (x, y, z) coordinates, giving $2 \times 21 \times 3 = 126$ values.

Concatenating these yields a **258-dimensional keypoint vector per frame** ($132 + 126 = 258$). MediaPipe Holistic additionally provides 468 facial landmarks per frame, which were extracted and inspected during dataset development but were not included in the final feature vector used to train the recognition model, keeping the input representation compact and well suited to real-time inference.

4.2 Sequence Construction

Each video clip is converted into a sequence of per-frame 258-dimensional keypoint vectors, forming a structured temporal representation of the gesture. Sequences correspond to

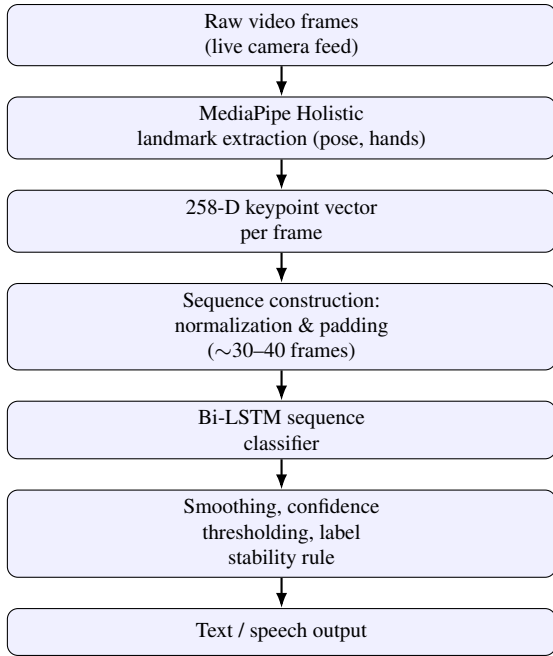


Figure 2: End-to-end data processing and inference pipeline, from raw video frames to text/speech output.

approximately 30–40 frames of motion. To ensure a consistent input shape across all training examples, sequences are normalized and padded to a fixed length, which is required for efficient batch training of the recurrent model. Figure 2 summarizes the full data processing pipeline, from raw video to model input.

4.3 Ineffective Baseline Models

Before arriving at the final architecture, two baseline approaches commonly used for static gesture recognition were evaluated and found inadequate for phrase-level recognition.

Static CNN classifiers. A convolutional neural network was trained to classify individual frames extracted from the video sequences. Because this model treats every frame independently, it cannot represent the temporal evolution that defines each gesture. As a result, gestures with similar static hand poses but different motion trajectories were frequently confused.

Frame-by-frame models. A second baseline produced a prediction for every frame and aggregated these per-frame predictions over time. This approach produced unstable and inconsistent outputs, since many dynamic gestures look nearly identical in any single isolated frame; without an explicit mechanism for modeling sequential dependencies, the resulting predictions exhibited high variability and low overall accuracy.

These baseline results motivated the adoption of a true sequence model capable of learning motion patterns over time.

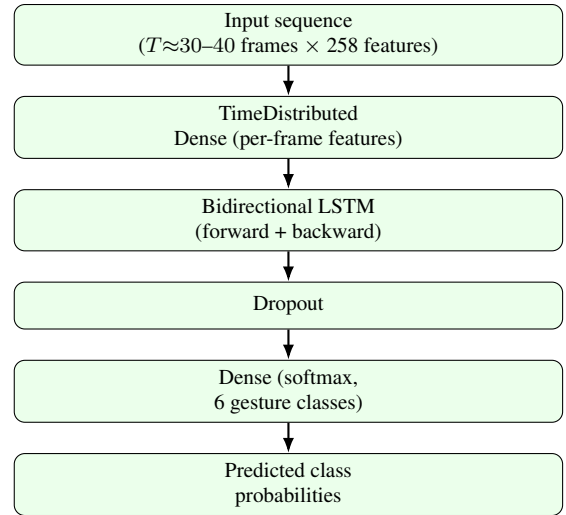


Figure 3: Bidirectional LSTM architecture used for phrase-level gesture classification. Class weighting is applied during training to address class imbalance.

4.4 Proposed Architecture: Bidirectional LSTM

The final model is a **Bidirectional Long Short-Term Memory (Bi-LSTM)** network that processes each 258-dimensional keypoint sequence in both the forward and backward directions, allowing it to capture long-range dependencies and subtle motion cues across the full duration of a gesture. The architecture incorporates:

- **TimeDistributed Dense layers**, which apply a consistent per-frame feature transformation before the sequence is passed to the recurrent layers;
- **Dropout layers**, which mitigate overfitting and improve generalization to unseen signers and recording conditions; and
- **Class weighting**, which compensates for imbalance across the six gesture categories during training.

The Bi-LSTM is trained on sequences of approximately 30–40 frames, each represented by the 258-dimensional MediaPipe-derived keypoint vector described in Section 4.1. Figure 3 illustrates the overall model architecture. This design proved substantially more stable and accurate than the frame-independent baselines of Section 4.3, and forms the foundation of the real-time recognition system described in Section 5.

5 Real-Time Inference Pipeline

Because the trained model is intended for live, on-device use, a recognition pipeline based purely on raw model output is insufficient: per-frame predictions can be noisy, and a single high-confidence frame should not, by itself, trigger a gesture output. To address this, the inference pipeline combines the Bi-LSTM model with several lightweight stability mech-

anisms:

- **Rolling sequence window:** A continuously updated buffer of the most recent frames is maintained, providing the temporal context the Bi-LSTM requires for each prediction.
- **Frame/temporal smoothing:** Predicted class probabilities are smoothed across consecutive frames to reduce frame-to-frame noise.
- **Confidence thresholding:** Predictions whose maximum class probability falls below a threshold are discarded, filtering out low-confidence or ambiguous frames.
- **Label-stability rule:** A gesture is only accepted once the same label has been predicted consistently for N consecutive frames, which substantially reduces false triggers from brief or accidental hand movements.

With these mechanisms in place, the pipeline maintains approximately **20–30 frames per second**, which is sufficient for smooth, low-latency inference during live use. Together, the rolling window, smoothing, thresholding, and stability rule ensure that the system responds primarily to deliberate, completed gestures rather than to incidental motion. This is an important property, since the *None* class (Section 3) was specifically designed to contain such incidental motion.

6 Experimental Setup

6.1 Dataset Splitting

The custom dataset was divided using a **stratified 70/15/15 train/validation/test split**, ensuring that each of the six gesture classes is proportionally represented in every subset. Stratification prevents the evaluation from being biased toward majority classes and provides a reliable basis for assessing generalization to unseen examples.

6.2 Training and Evaluation Metrics

The Bi-LSTM model (Section 4.4) was trained with class weighting to mitigate the effects of class imbalance in the custom dataset. Training and validation accuracy and loss were monitored throughout training to track learning progress and to detect overfitting or underfitting. The following metrics were used to characterize the trained model:

- **Training, validation, and test accuracy**, to measure overall correctness at each stage;
- **Loss curves**, to monitor convergence and detect overfitting;
- **Test-set accuracy**, used as the primary measure of generalization to unseen data; and
- **Per-class precision, recall, F1-score, and a confusion matrix**, used to identify which gesture classes were most commonly confused with one another.

In addition to these offline metrics, the system was evaluated for **real-time responsiveness** (frames per second) and **prediction stability** under the rolling-window, smoothing, thresholding, and label-stability mechanisms described in Section 5.

Table 1: Per-class classification metrics on the validation split ($n = 305$).

Class	Precision	Recall	F1-score	Support
Hello	0.75	0.84	0.79	45
Peace	0.84	0.83	0.83	58
Thank You	0.82	0.77	0.80	71
Yes	0.82	0.75	0.79	44
None	0.83	0.87	0.85	46
No	0.79	0.80	0.80	41
Accuracy			0.81	305
Macro avg	0.81	0.81	0.81	305
Weighted avg	0.81	0.81	0.81	305

7 Results

7.1 Learning Curves

Figure 4 shows training and validation accuracy and loss over the course of training. Training accuracy increases steadily and approaches 1.0, while validation accuracy rises more gradually and stabilizes in the mid-to-high 0.80s. Correspondingly, training loss decreases monotonically, while validation loss decreases initially and then levels off, with a small remaining gap between training and validation curves that is consistent with the regularization provided by dropout and class weighting.

7.2 Validation Performance

Table 1 reports per-class precision, recall, and F1-score on the validation split ($n = 305$), together with macro- and weighted-average metrics. The overall validation accuracy is **81%**, with broadly balanced performance across all six classes: per-class F1-scores range from 0.79 (*Hello* and *Yes*) to 0.85 (*None*).

Figure 5 shows the corresponding validation confusion matrix. The diagonal dominance indicates that the model correctly classifies the large majority of examples in every class. The most notable confusion is between *Thank You* and *Hello* (8 examples) and between *Thank You* and *None* (6 examples), which is consistent with these gestures sharing similar starting hand positions near the face or chest before diverging in their motion trajectories.

7.3 Test Accuracy and Real-Time Performance

On the held-out test split, the final Bi-LSTM model achieves an overall **test accuracy of 85.57%** across all six gesture classes, reflecting strong generalization beyond the validation set. In live operation, the full pipeline, including MediaPipe Holistic landmark extraction, the Bi-LSTM classifier, and the smoothing/stability layer described in Section 5, sustains approximately **20–30 frames per second**, confirming that the system is suitable for real-time, on-device use.

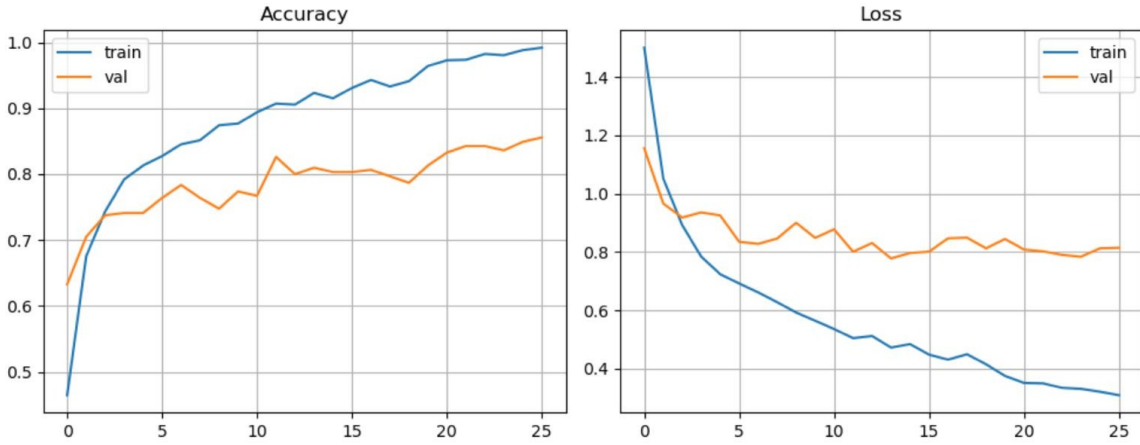


Figure 4: Training and validation accuracy (left) and loss (right) over the course of training.

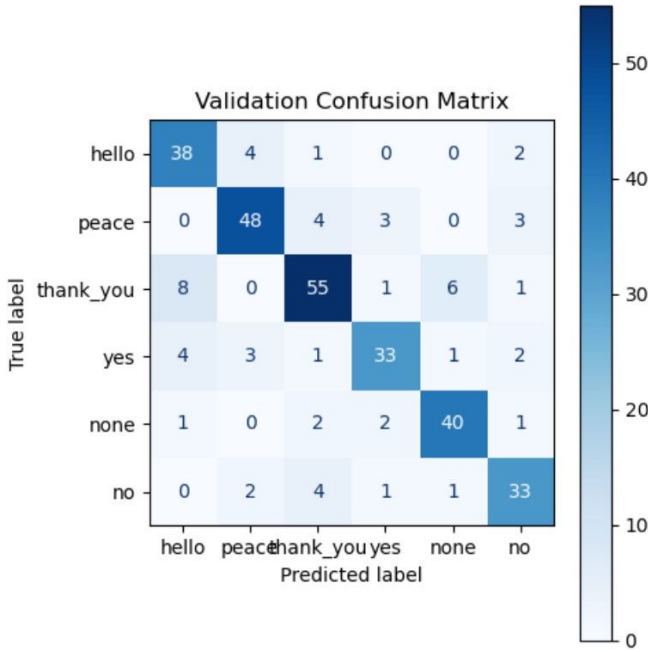


Figure 5: Validation confusion matrix across the six gesture classes ($n = 305$).

8 Discussion

8.1 Strengths

Within the scope of the six defined gesture classes, the system exhibits several practical strengths. The consistent, well-structured keypoint sequences produced by MediaPipe Holistic provide a compact and informative input representation, supporting accurate classification. The deliberately diverse *None* class and the multi-stage smoothing pipeline (rolling window, temporal smoothing, confidence thresholding, and label-stability rule) substantially reduce false positives during real-time use. Finally, the system maintains real-time frame rates (20–30 FPS) with low-latency predictions, and the label-stability rule, which requires multiple consecutive

consistent predictions before a gesture is confirmed, further improves reliability in live interaction.

8.2 Limitations

Several limitations define the current scope of the system. First, the model supports only **six gesture classes**, which limits its expressiveness for general communication. Second, the system depends on **clear upper-body visibility**; occluded or partially visible hands degrade the quality of the extracted keypoints and, in turn, classification accuracy. Third, **partial or incomplete gestures**, which lack the full temporal signature the Bi-LSTM was trained on, may be misclassified. Finally, real-time accuracy can be affected by **environmental factors** such as poor lighting, low-quality cameras, or limited on-device compute capacity.

8.3 Use of AI Assistance

In keeping with current norms for disclosing the use of AI tools in research, we note that an AI assistant was used as a collaborative aid throughout this project. Specifically, AI assistance contributed to: automating the conversion of raw video clips into structured numerical sequences, ensuring uniform preprocessing across all samples; suggesting candidate model architectures, learning rates, sequence lengths, and dropout values during hyperparameter exploration; identifying and correcting bugs in preprocessing, data-loading, and model scripts; refining gesture definitions and data-collection instructions for consistency across team members; and tuning the smoothing functions, confidence thresholds, and label-stability rules used in the real-time pipeline. All modeling decisions, dataset design choices, and reported results were reviewed and validated by the authors.

9 Conclusion and Future Work

This paper presented a real-time, on-device system for phrase-level sign language recognition. The system com-

bines a custom 1,500-clip dynamic-gesture dataset, including a deliberately robust *None* class, with a Bidirectional LSTM classifier operating on 258-dimensional MediaPipe Holistic keypoint sequences, and a rolling-window inference pipeline with smoothing, confidence thresholding, and a label-stability rule. The resulting system achieves 85.57% test accuracy across six gesture classes while sustaining 20–30 FPS in live operation, demonstrating that lightweight, keypoint-based sequence models are a practical foundation for accessible, real-time gesture-to-text and gesture-to-speech translation.

Several directions for future work follow naturally from the limitations discussed in Section 8:

- **Expanding the gesture vocabulary**, including a move toward sentence-level recognition of multi-sign phrases rather than a fixed set of six classes;
- **Increasing dataset diversity**, by incorporating more participants, recording environments, and motion variations;
- **Adopting transformer-based sequence models**, which may offer improved temporal modeling capacity for longer or more complex gesture sequences;
- **Developing a mobile version** of the pipeline, optimized for on-device inference on phones and wearables to broaden accessibility;
- **Deploying on smart glasses**, so that the recognition pipeline can run directly on a head-mounted camera and provide hands-free, in-the-moment gesture-to-text and gesture-to-speech translation. Such a device would let a sign language user communicate naturally with people who do not know sign language, with the glasses translating the user’s signed phrases into spoken or written language in real time so that the other person can understand them, helping individuals with speech and hearing impairments take part in everyday conversations more easily; and
- **Supporting multilingual speech output**, so that recognized gestures can be rendered as spoken language across a range of target languages.

References

- [1] D. Li, C. Rodriguez, X. Yu, and H. Li, “Word-level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2020.
- [2] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [3] M. Schuster and K. K. Paliwal, “Bidirectional Recurrent Neural Networks,” *IEEE Transactions on Signal Processing*, vol. 45, no. 11, pp. 2673–2681, 1997.
- [4] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong,

J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann, “MediaPipe: A Framework for Building Perception Pipelines,” *arXiv preprint arXiv:1906.08172*, 2019.

- [5] N. C. Camgoz, S. Hadfield, O. Koller, H. Ney, and R. Bowden, “Neural Sign Language Translation,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.